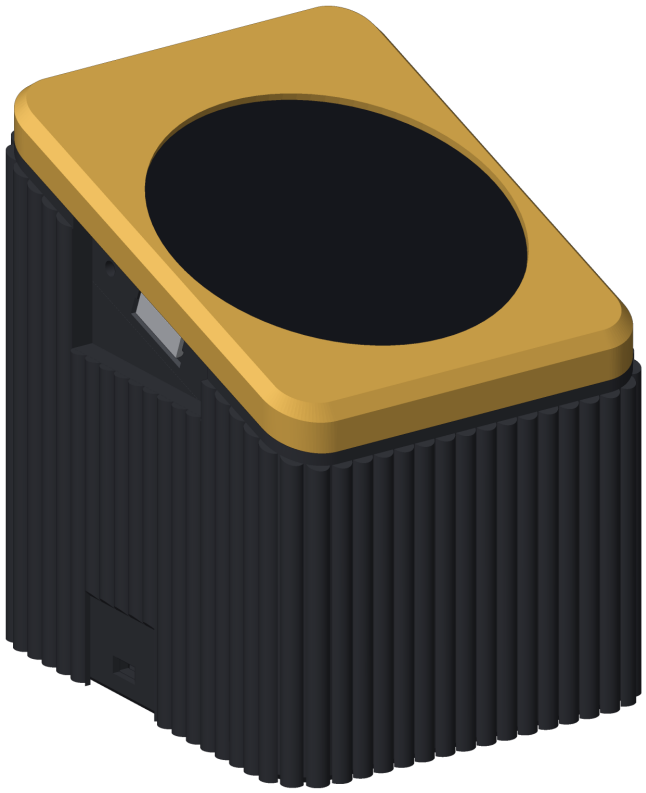
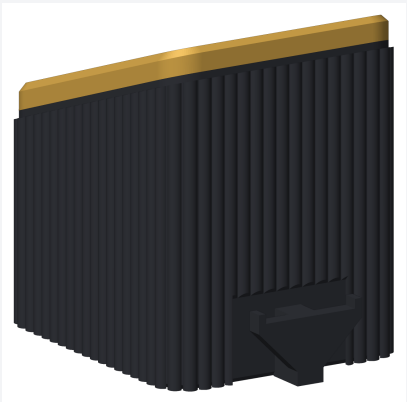
LyricFlash Case

Assembly guide

For the Waveshare ESP32-S3-Touch-AMOLED-1.75 · 20° tilted desk case · design v10.4
9 steps · 5 screws, 8 magnets · no glue





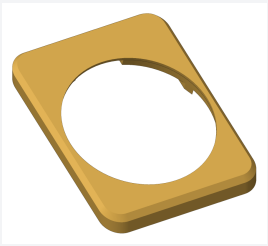
How it goes together

- Magnets, the switch and the battery go into the black base.
- The display board screws onto a printed carrier plate, on the bench.
- Board and plate go into the base USB-C first; two screws hold the plate down.
- The gold shell goes on last, held by the magnets. The cable turns round the cleat.

What you need

Parts, hardware, tools and print settings

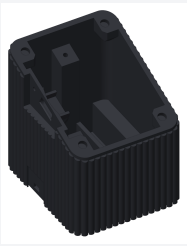
Printed parts



Top shell

Gold PLA Metal · ×1

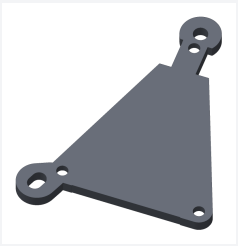
Holds the screen, 4 magnets



Base

Black PETG · ×1

Ribbed, panels or smooth; with or without the cleat

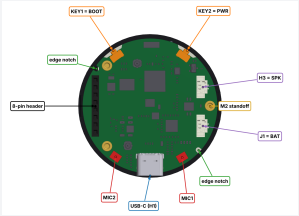


Carrier plate

Black PETG · ×1

The board screws to it

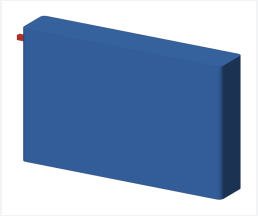
Electronics



Display board

Waveshare ESP32-S3-Touch-AMOLED-1.75

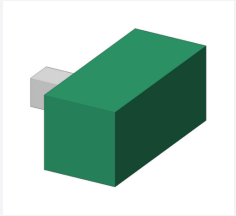
Battery plug J1



LiPo battery

3.7 V 2000 mAh · 56 × 34.5 × 10.3 mm

With the plug that fits J1



Slide switch

SS-12D00 (optional)

Hard power-off for storage

Hardware

- 3× M2 × 3 flat head (countersunk): board to the plate
- 2× M2.5 × 6 pan head: plate to the base (they cut their own thread, no nuts)
- 8× 4 × 2 mm disc magnets: 4 in the base, 4 in the shell

Tools

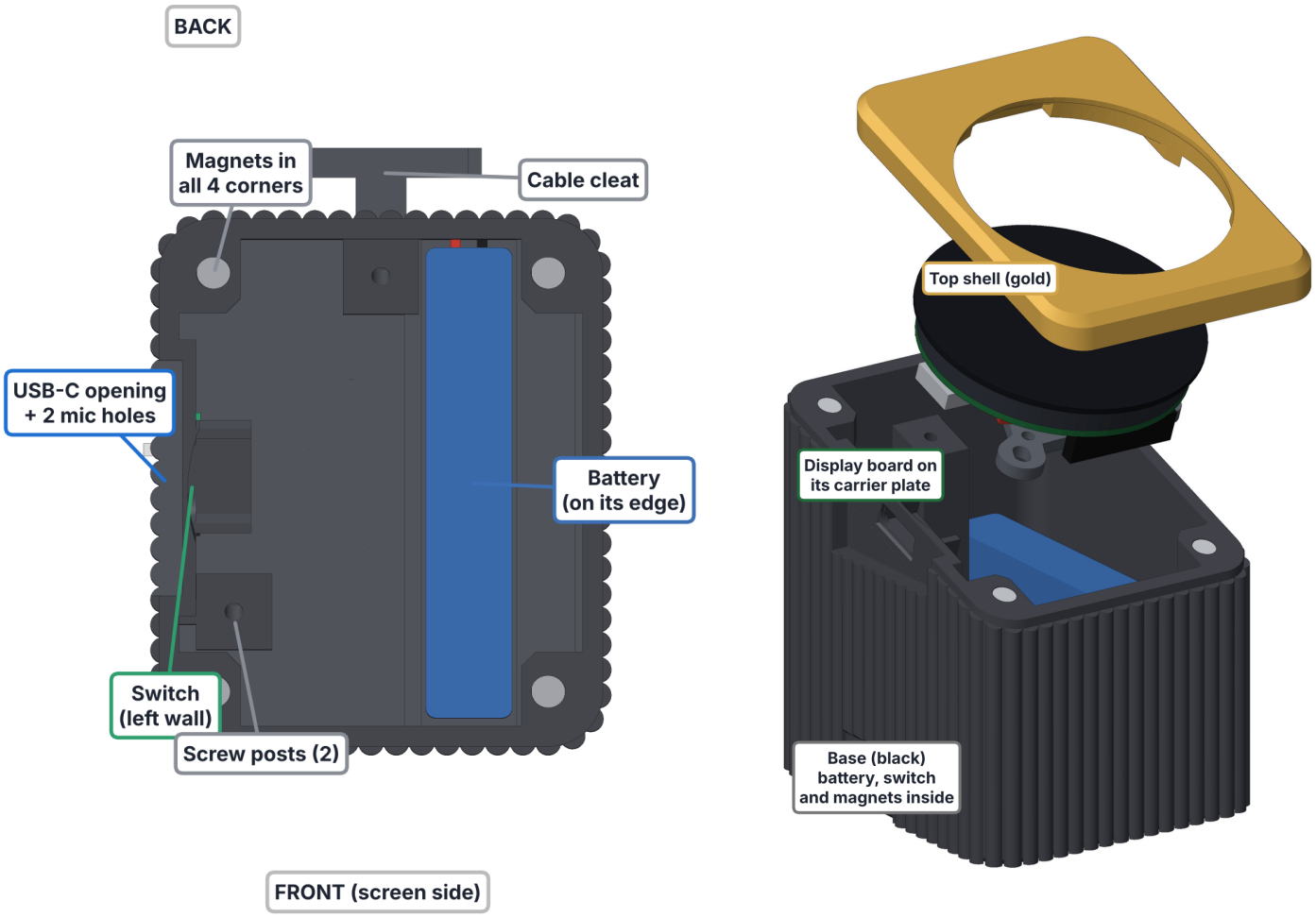
- Soft cloth (the screen goes face-down on it)
- Small Phillips / hex driver for M2 and M2.5
- Multimeter (battery polarity check)
- Flush cutters (support clean-up)
- Only for the switch: soldering iron and heat-shrink

Print settings

- 0.2 mm layers, 4 walls, 15 % gyroid, 0.4 mm nozzle
- Top shell: face-down (screen ring on the plate), no supports
- Base: floor down, supports off. Paint a support enforcer on the top edge of the USB-C opening only.
- Carrier plate: flat, board side down (countersinks facing up), no supports

Where everything goes

Base seen from above, and the three layers



Three layers: the gold top shell, the display board on its carrier plate, and the black base. The battery stands on its edge right of centre. The switch is in the left wall under the USB-C port and the two microphone holes. Magnets sit in all four corners, and the plate screws to two posts, front-left and back-centre.

Before you start

Five quick checks

Clean the USB-C opening

Remove all the painted support from inside the opening. Hold the bare board's USB-C socket up to it: it should slide in freely.

Check the two mic holes

Two Ø2 mm holes, one either side of the USB-C opening. They must be clear all the way through. LyricFlash listens through them.

Check the battery polarity

Use a meter before you plug the battery in. Red/black order isn't the same from every seller, and a reversed LiPo can kill the board's power chip.

Know what isn't reachable

There's no access to the PWR or BOOT buttons. The board powers up from USB or the battery, and the switch turns it off. To flash in recovery mode, open the case.

Never press on the screen

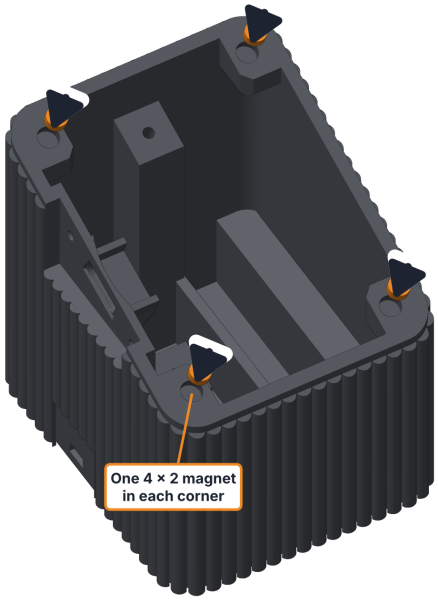
Every push and every screw in this guide goes plastic-to-plastic, outside the screen. The shell's lip stands clear of the glass, so even the magnets don't press on it.

Which base? All six bases (ribbed, panels or smooth, each with or without the cleat) assemble the same way. Bases without the cleat skip step 9.

Into the base

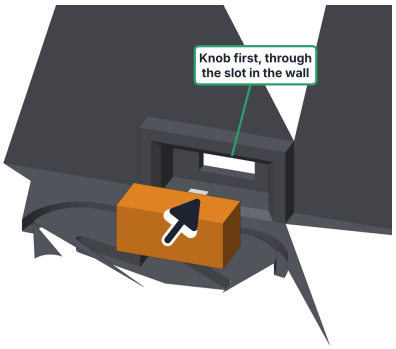
1 Magnets

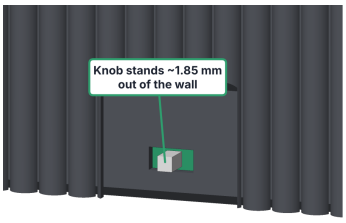
- Press one 4 × 2 magnet into each corner pocket of the base, flush with the top. Do the same in the four corner pockets on the underside of the gold shell.
- **Check polarity first: stack each shell magnet on its base magnet so they attract, mark the faces that meet, and press them in that way round.**
- A snug press-fit. A drop of CA glue if one is loose.



2 Slide switch (optional)

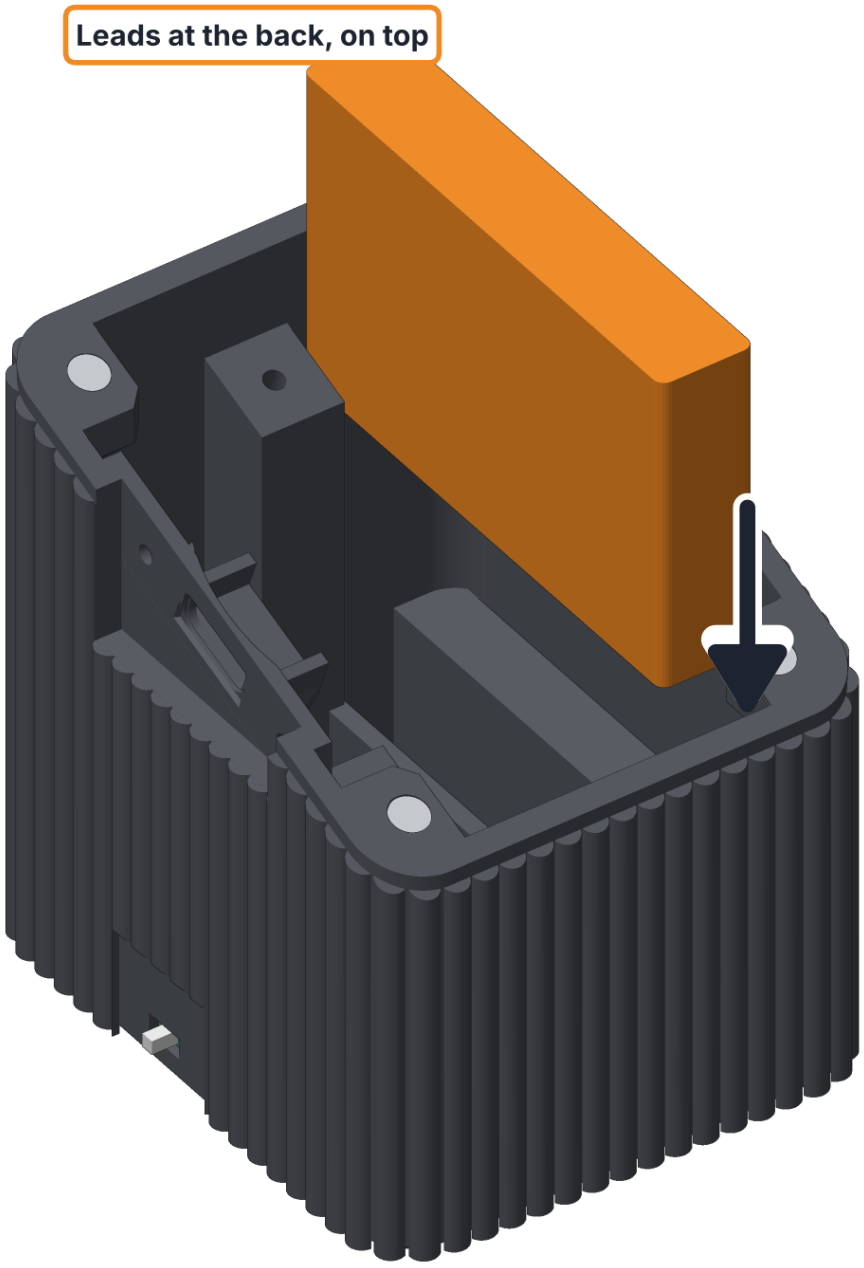
- Solder the switch into the battery's red (+) lead. Insulate both joints.
- From inside the case, push it into its pocket in the left wall under the USB-C, knob first through the slot.
- It's a firm press-fit, which is meant to be. The knob ends up about 1.85 mm out of the wall and slides end to end.
- Not fitting a switch? Skip to step 3.





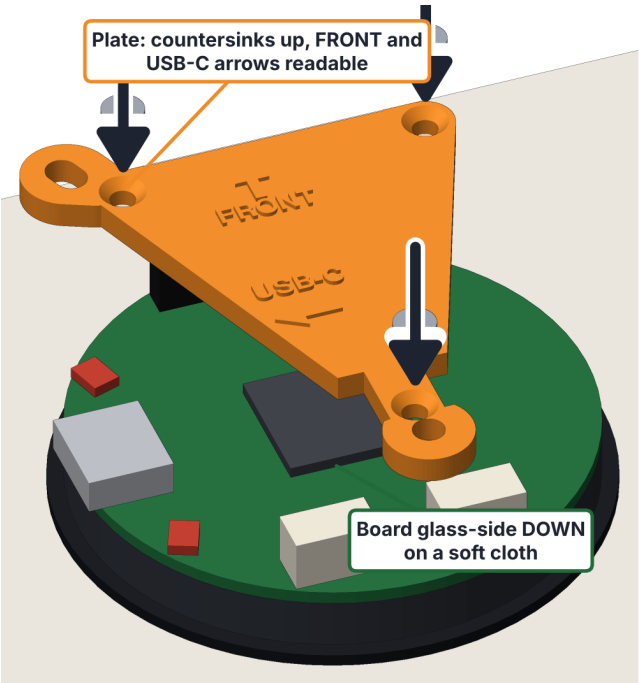
3 Battery

- Stand the battery on its thin edge and lower it straight down into its slot, right of centre.
- The leads go at the back end, at the top.
- Nothing hangs over the slot, so it goes in without tilting. It should drop in by hand, with no forcing.



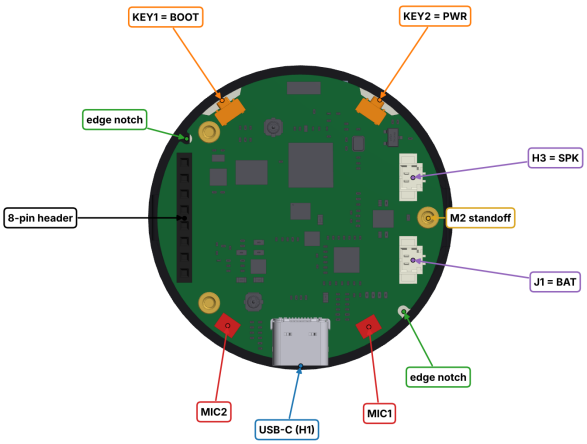
4 Screw the board to the carrier plate (on the bench)

- Lay the board glass-side down on a soft cloth.
- Put the carrier plate on its back, countersinks facing up. The raised FRONT and USB-C arrows should read correctly and point to the board's front and its USB-C. If you see them mirrored, flip the plate.
- Line the holes up over the board's three metal standoffs.
- Fit the three M2 × 3 flat heads. Snug them, don't crank them: they go into the board's own standoffs.



5 Plug in the battery

- Plug the battery into J1 on the back of the board.
- **Check the battery polarity with a meter first (see Before you start).**

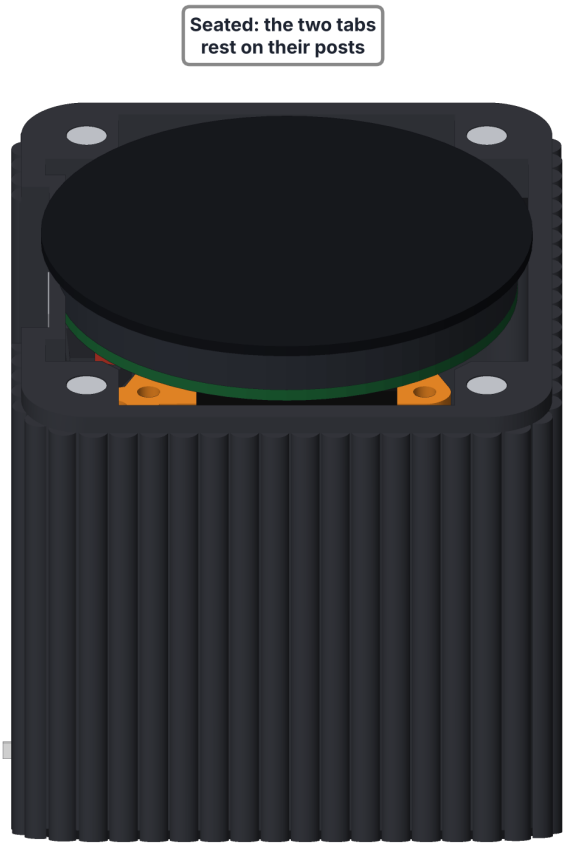


Board into the base

6 USB-C first, then lower it down

- Hold the board by its plate, glass up, tilted.
- Put the board's USB-C socket into its opening in the left wall first.
- Then lower the right side down. The two plate tabs land on their posts. Nothing should rub on the way.
- Keep the battery wires clear so they don't get pinched under the plate.



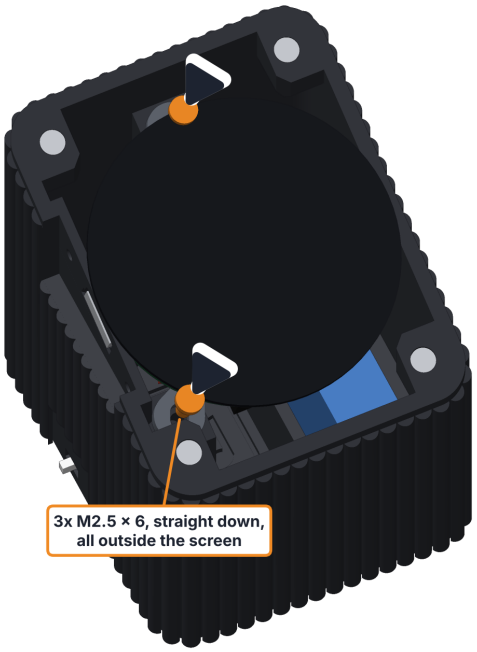


If it doesn't settle by itself, don't push on the screen. Lift it out and look for a pinched wire.

Fix it and close it

7 Screw the plate down

- Two M2.5 × 6 pan heads, straight down through the plate's tabs into the posts, both outside the screen.
- Back-centre first (round hole). Then the front-left one: its hole is a short slot, so it always finds its post.
- They cut their own thread the first time, so it's a little stiff. Stop as soon as the head seats.
- No nuts needed.



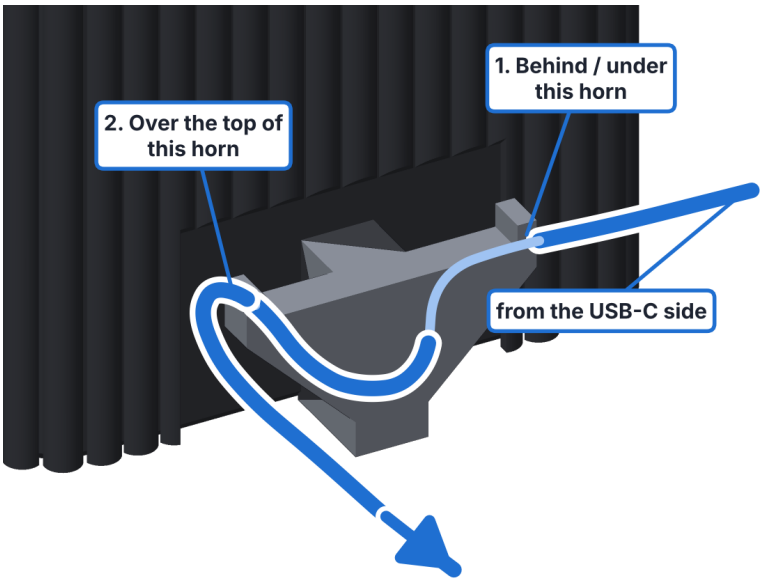
8 Gold shell on

- Put the shell straight down from above, square to the screen.
- The rim under the shell drops inside the base walls and lines it up. The magnets pull it home with a click.
- The lip stands 0.25 mm clear of the glass, so nothing presses on the screen.
- Check: the seam is even all round, with no rattle.



9 Cable cleat

- Plug in the USB-C cable and bring it round to the back.
- Give it one turn round the cleat at the bottom of the back wall: under one horn, over the other.
- A tug on the cable now pulls on the cleat, not the USB-C plug.
- No cleat on your base? Nothing to do.



Opening the case later

- Pull the gold shell straight up off the magnets.
- Undo the two M2.5 screws. Lift the board and plate out, right side first, unhooking the USB-C last.
- Lift the battery straight out.